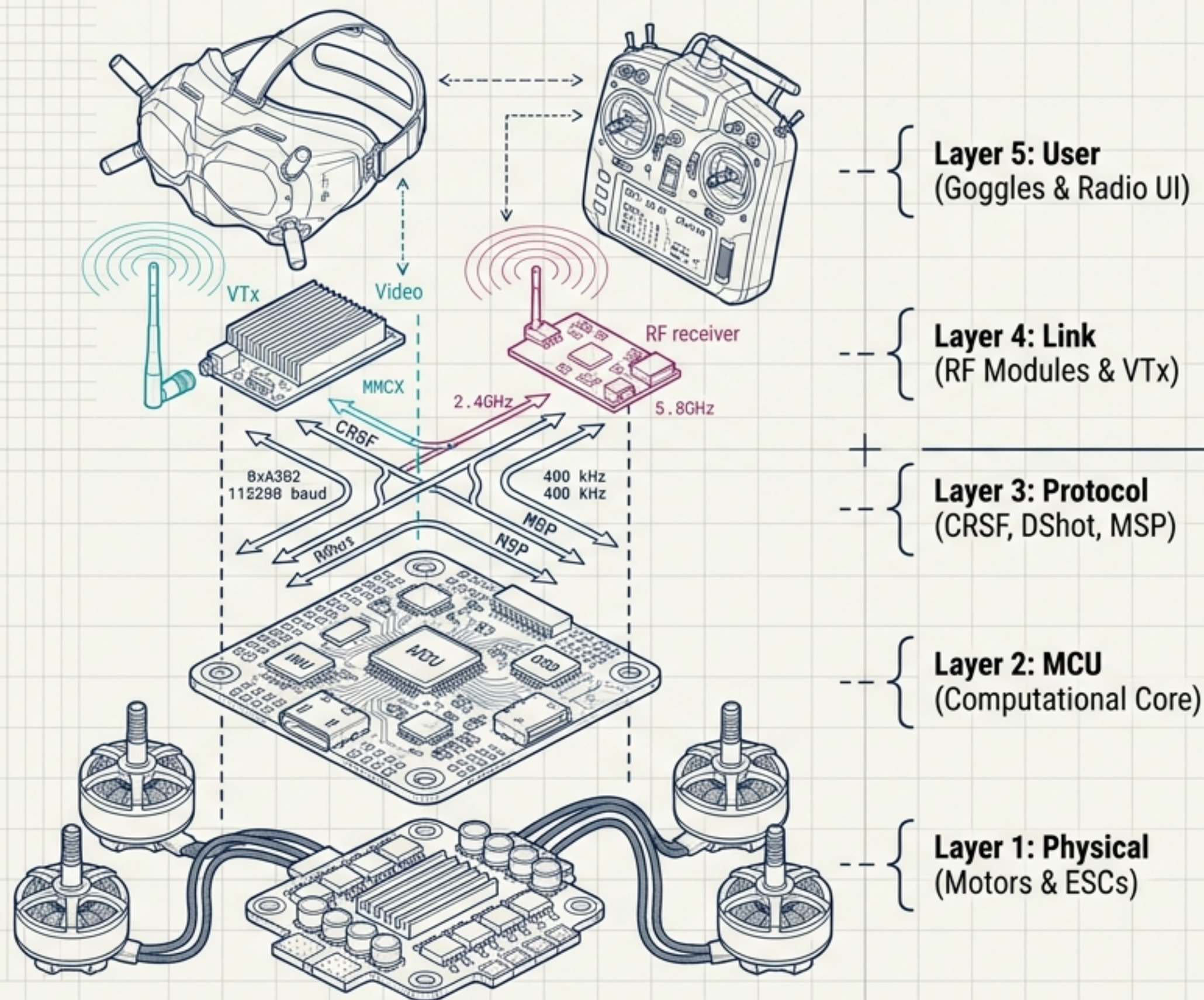


EXECUTIVE OVERVIEW: THE FPV SOFTWARE STACK



REAL-TIME EMBEDDED ECOSYSTEM

Microcontroller architectures bridging high-frequency RF engineering with advanced closed-loop control theory.

BIDIRECTIONAL INTEGRATION

The definitive shift away from one-way pulse-width signals (PWM/PPM) to unified, bidirectional digital telemetry.

THE MASTERCLASS SCOPE

Deconstructing the computational core, communication protocols, rendering pipelines, and dynamic filtering systems defining the modern FPV architecture.

FLIGHT CONTROLLER FIRMWARE ARCHITECTURE

Diagnostic Note: Firmware designates the MCU's mission profile—transforming identical silicon into either an ultra-responsive race platform or a completely autonomous drone.



BETAFLIGHT

[Raw Performance]

Architecture:

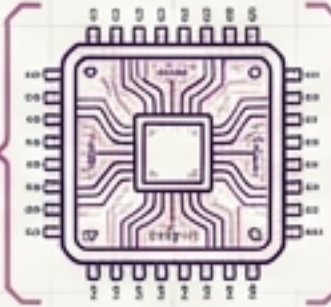
Loop rates up to 8k/8k, custom PID tuning, Blackbox data logging.

Weakness:

Basic GPS (Rescue mode only, no waypoints/ALTHOLD).

Best For:

Racing, Freestyle, manual quads requiring zero-latency response.



iNAV

[Autonomous Navigation]

Architecture:

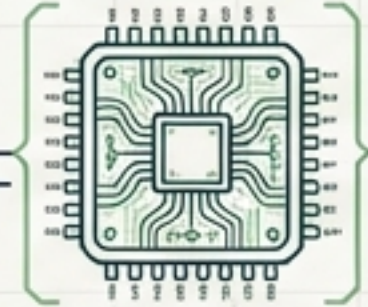
GPS-assisted workflows, WAYPOINT missions, ALTHOLD, advanced Return-to-Home (RTH) with wind compensation.

Weakness:

Lacks deep, complex mission logic.

Best For:

Fixed-wing aircraft, long-range FPV cruising.



ARDUPILOT

[Deep Autonomy]

Architecture:

Advanced mission logic, complex mapping, VTOL support, robust failsafes.

Weakness:

Extreme setup complexity and parameter management overhead.

Best For:

Professional UAV work, academic research, complex multirotors/rovers.

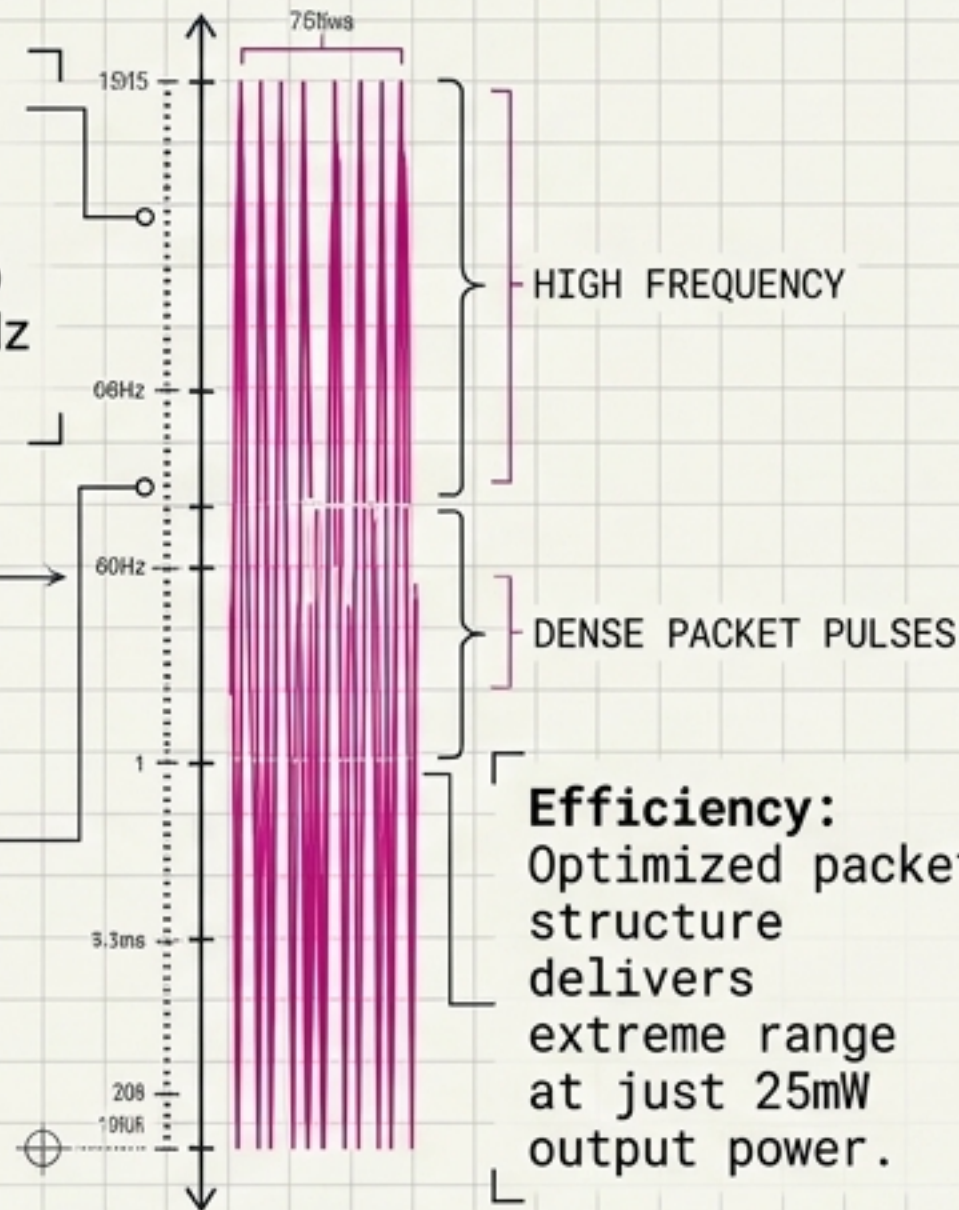
HIGH-PERFORMANCE RADIO LINK PROTOCOLS

EXPRESSLRS (ELRS)

Open-source LoRa modulation.

Bands: 2.4GHz (up to 500Hz) & 900MHz (up to 200Hz).

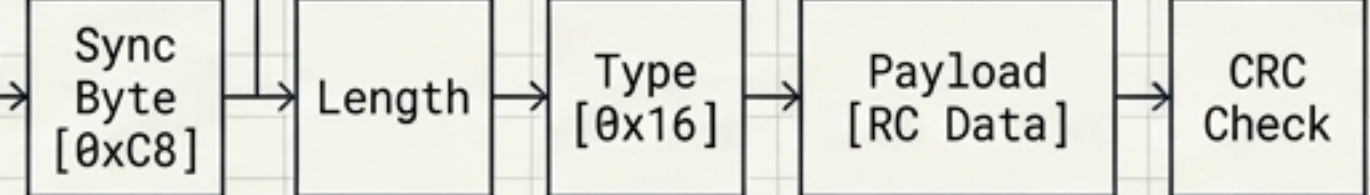
Latency: 3-5ms ultra-low end-to-end response.



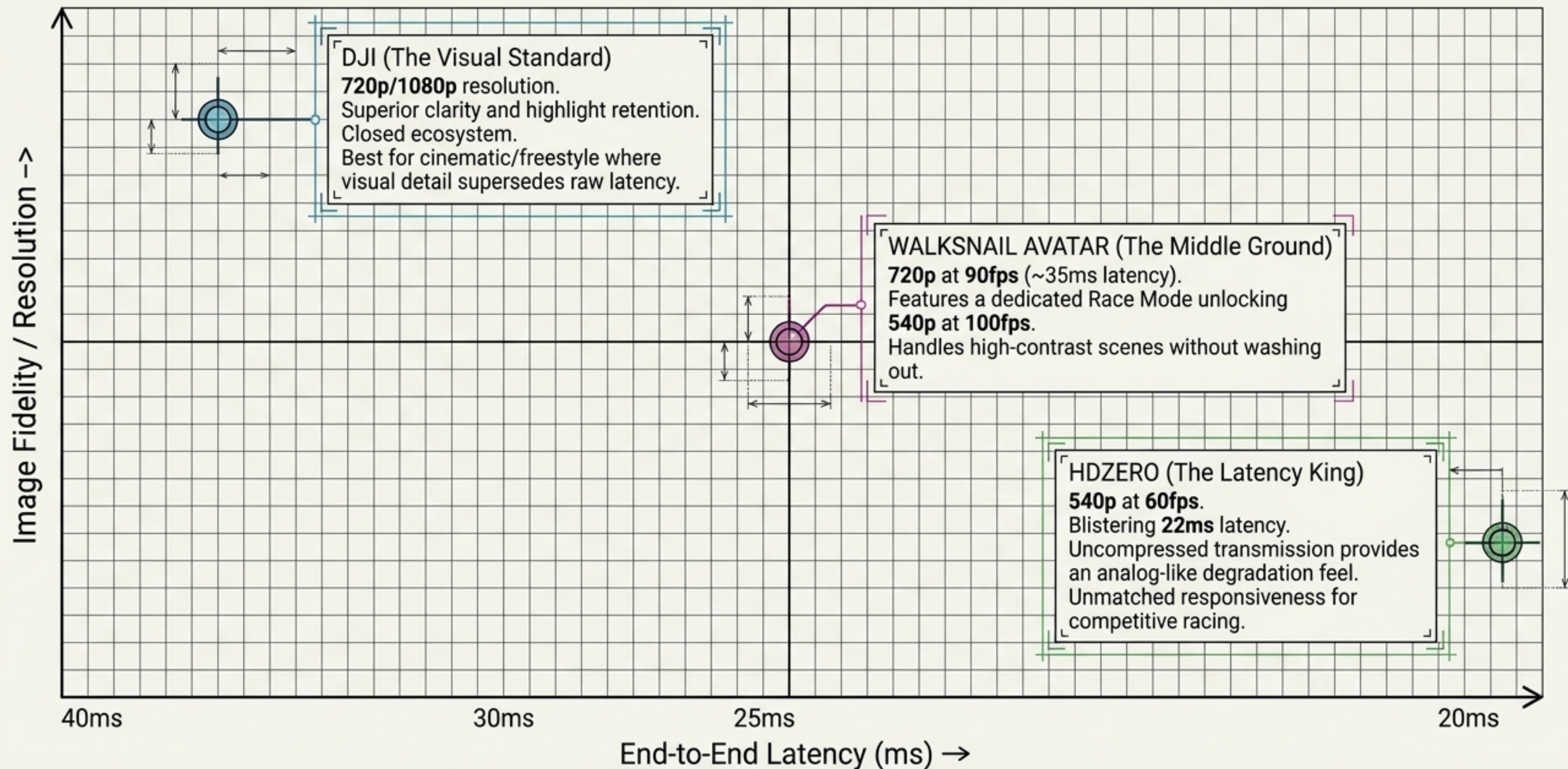
TBS CROSSFIRE / CRSF

Proprietary 900MHz ecosystem. Up to 150Hz. Premium hardware supporting up to 2W telemetry power for extreme-distance links (20-30km).

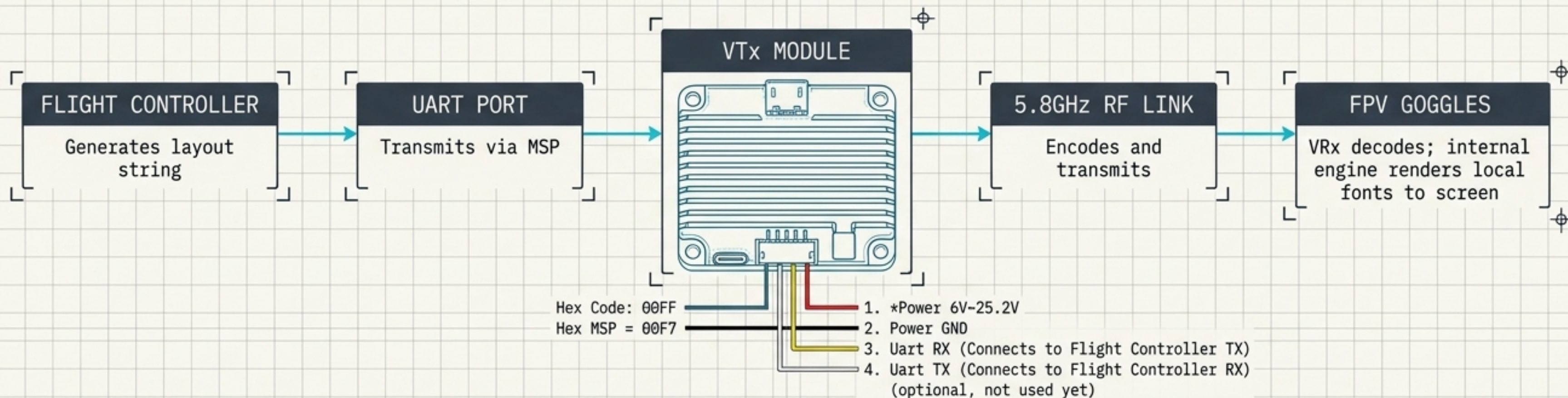
CRSF Protocol Architecture (420kbps Bandwidth)



DIGITAL VIDEO TRANSMISSION ECOSYSTEMS



OSD SYSTEMS AND THE MSP DISPLAYPORT PIPELINE



Legacy Telemetry OSD (Protocol 32)

Goggles utilize their own internal rendering engine to display raw sensor data. The flight controller has zero control over visual styling or placement.

MSP DisplayPort / Canvas Mode (Protocol 42/33)

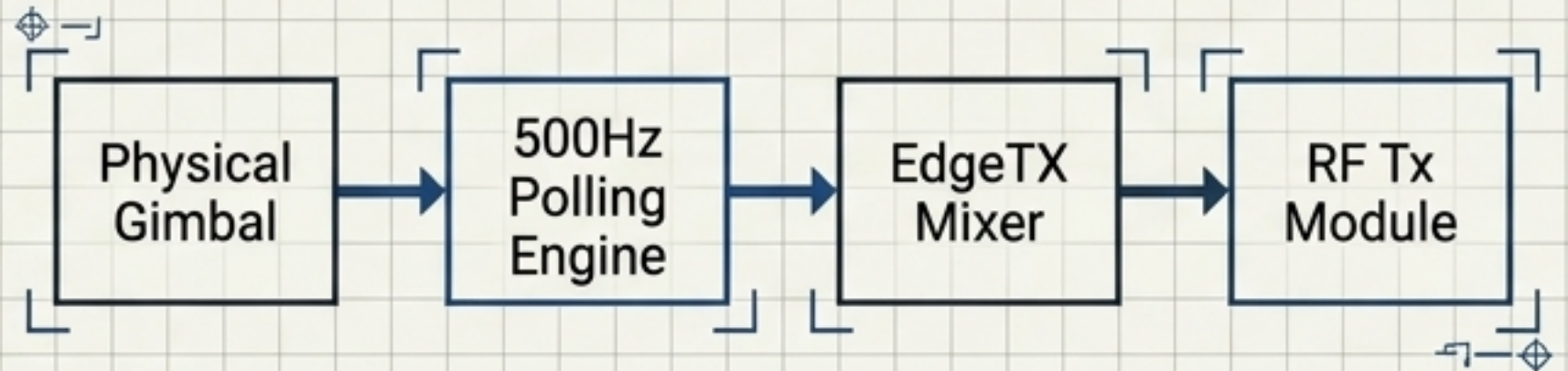
Operates as a remote character-based framebuffer. The Flight Controller dictates the exact layout and position, while the OSD hardware provides the local typography to render the strings.

Implementations: Native on HDZero & Walksnail. Enabled on DJI hardware via community wtf-os and msdp-osd modules.

TRANSMITTER FIRMWARE: EDGETX & LUA SCRIPTING

EDGETX ARCHITECTURE

- A highly optimized fork of OpenTX designed for modern hardware architectures.
- Features an intuitive Touchscreen UI and 'OneBit Mode' (a software inverter for legacy FrSky hardware).
- The 500Hz Advantage: Features crucial 500Hz Gimbal Polling, which eliminates internal radio processing latency when utilizing 1000Hz ELRS RF links.



LUA SCRIPTING BRIDGE

Local Lua scripts run **directly** on the transmitter, utilizing **bidirectional CRSF/ELRS telemetry** to execute over-the-air FC parameter changes (PIDs, VTx channels) without a physical USB connection.

```
local function run(event)
  -- Request PID values via CRSF telemetry
  crossfireTelemetryPush(0x16, {0x01, 0x0A})

  -- Render VTX channel to EdgeTX UI
  lcd.drawText(10, 20, 'VTX CH: ' .. vtx_band, 0)
end
```


FPV SIMULATORS & SIM-TO-REAL TRANSFER

VELOCIDRONE

(Code-Level Simulation)

- Pure racing focus.
- **Computational Approach:** Merges proprietary aerodynamics engines directly with raw **Betaflight source code**.
- **Outcome:** Ensures exact **PID loop** and rate translation identical to real-world embedded systems.
- Features high-performance, low-latency **"Nemesis Mode"** multiplayer tracking.

LIFTOFF

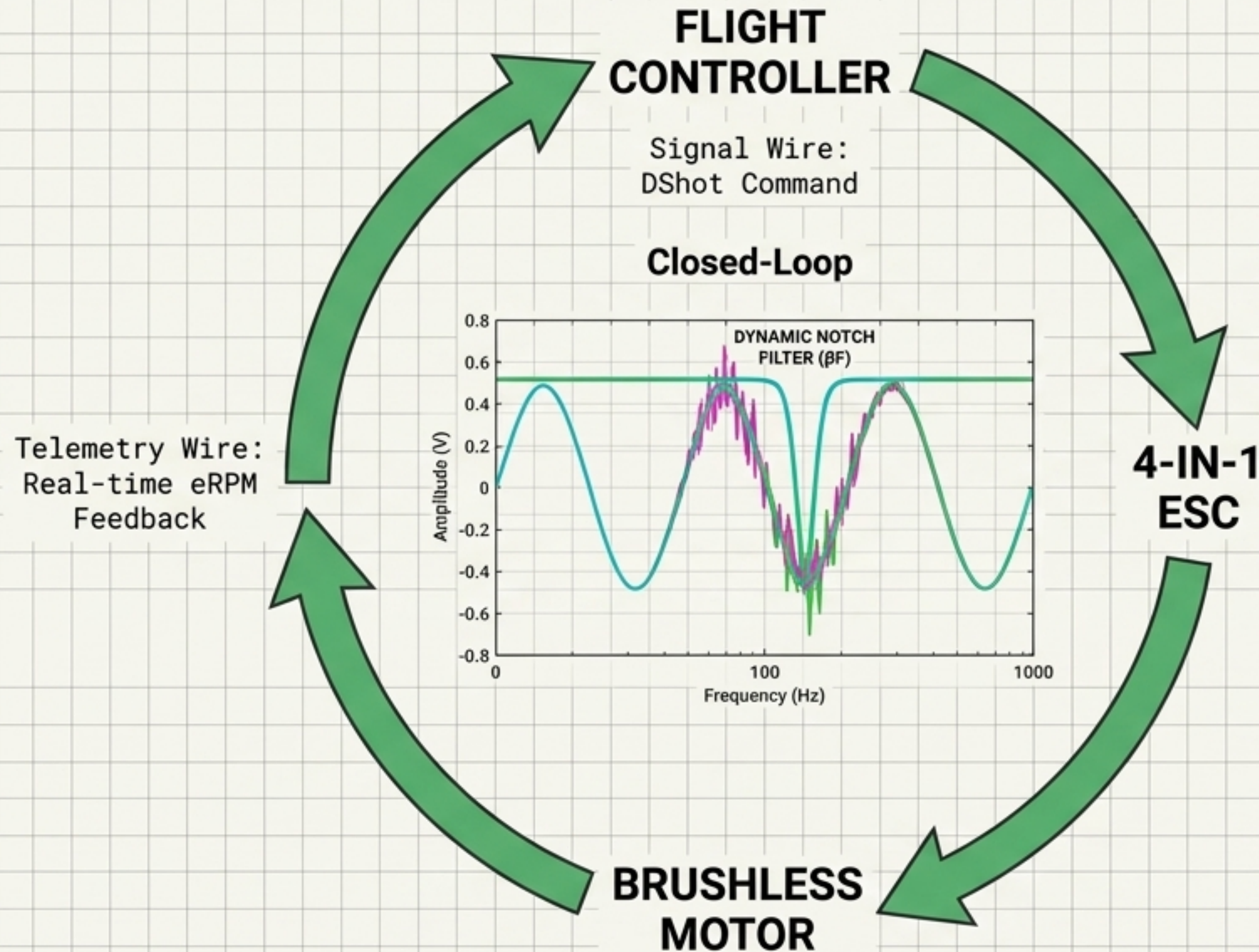
(Component-Level Simulation)

- Freestyle and cinematic focus.
- **Computational Approach:** Built on a **Unity-based** physics engine.
- **Outcome:** Utilizes a deep **"Parts System"** that calculates unique weight, thrust, and aerodynamic drag profiles dynamically based on the **3D models** of specific virtual components.

THE SIM-TO-REAL TRANSFER RULE

True mastery and muscle memory translation relies absolutely on utilizing **identical physical controller hardware**. Furthermore, **USB polling rates** must perfectly mirror the live embedded **FC loop latency** to ensure flawless neurological adaptation.

RPM FILTERING VIA BIDIRECTIONAL DSHOT



THE NOISE PROBLEM

Mechanical frame resonance and motor vibrations overwhelm the sensitive gyro sensor. Legacy static low-pass filters solve this but introduce massive processing phase delay.

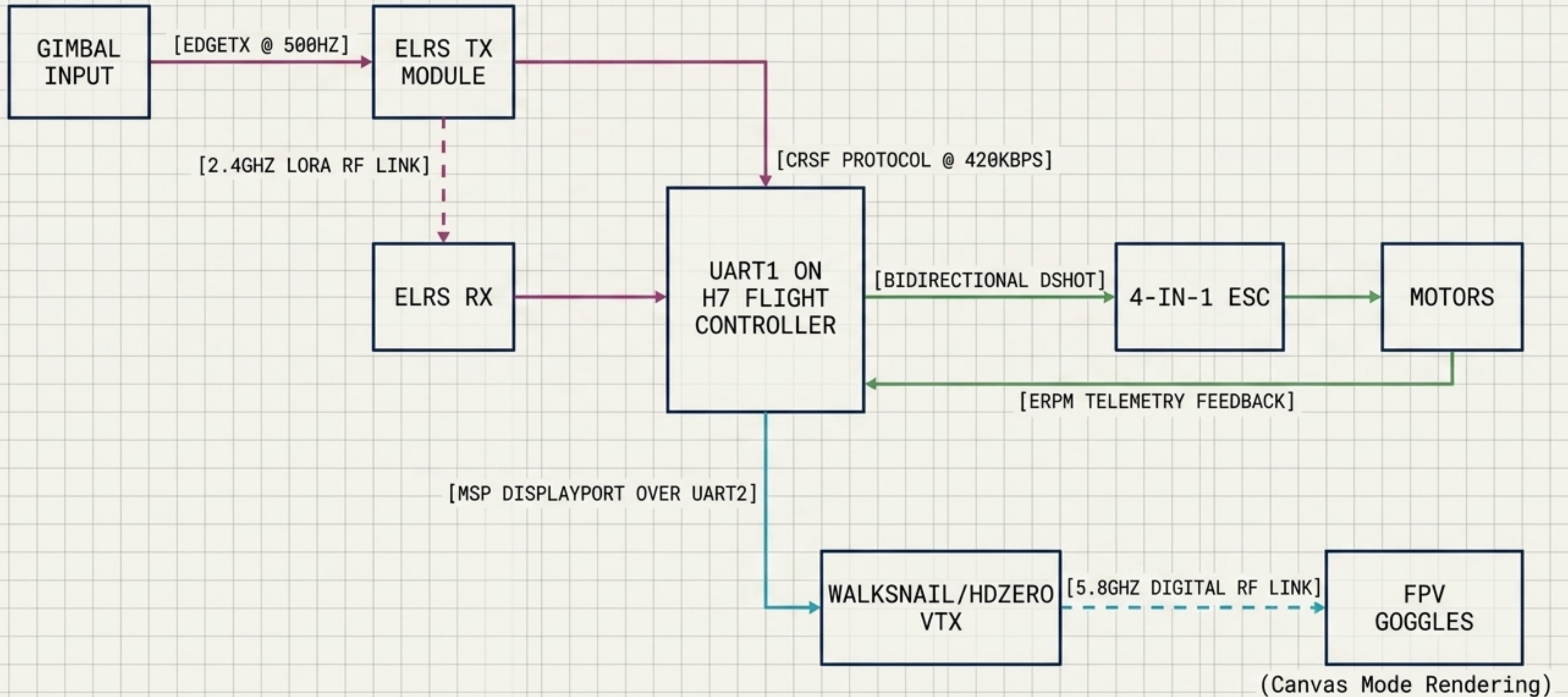
THE DSHOT SOLUTION

Bidirectional DShot allows the ESC to transmit precise eRPM (electrical RPM) data back to the Flight Controller over the signal wire in real-time.

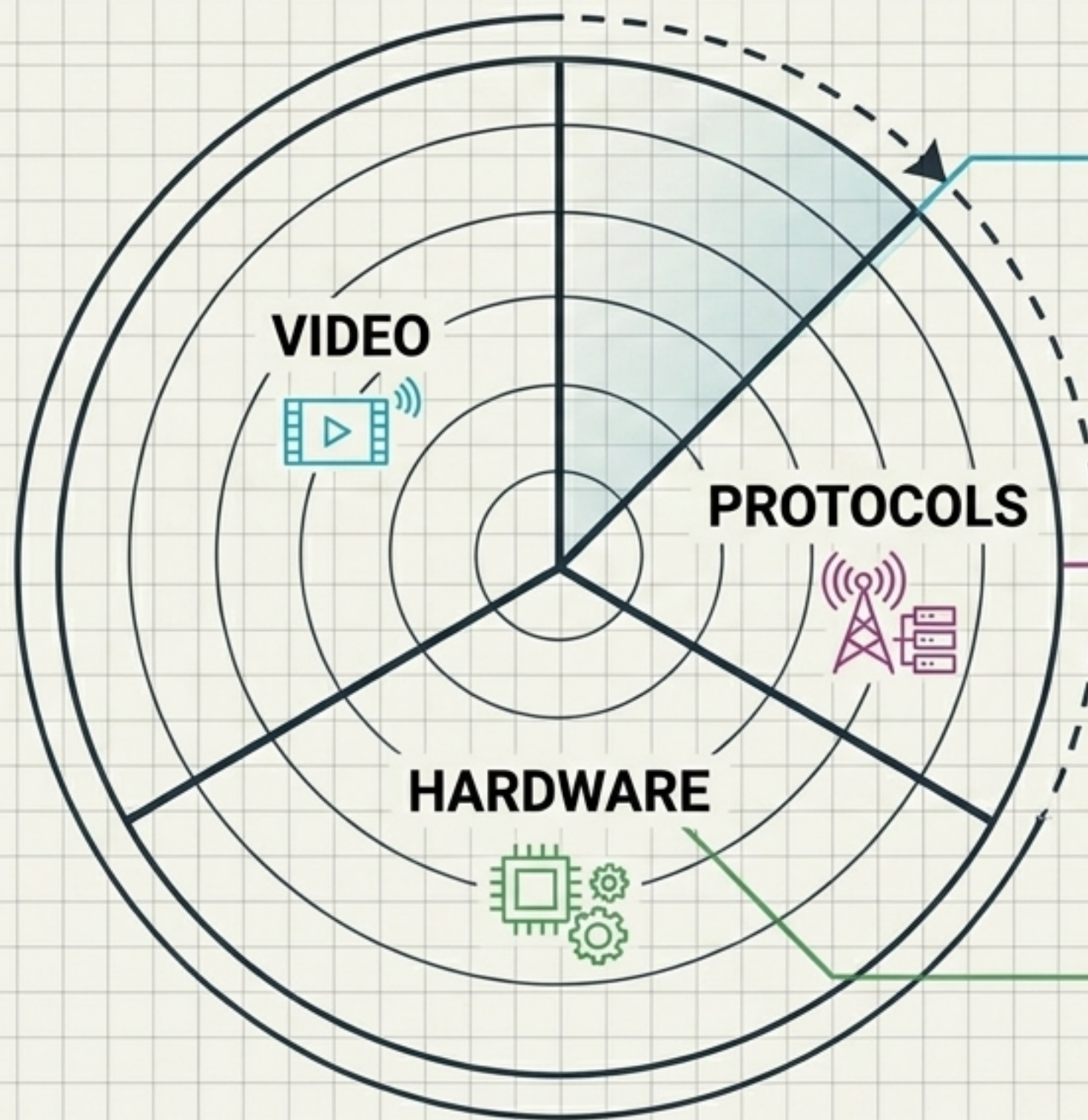
DYNAMIC NOTCH FILTERING

Betaflight consumes this eRPM telemetry to dynamically calculate and slide digital 'notch filters' directly over the exact noise frequencies generated by the motors. This frees up MCU bandwidth and dramatically lowers control latency.

MASTER SYNTHESIS: FULL UAV SYSTEM INTEGRATION SCHEMATIC



HORIZON: FUTURE TRENDS IN FPV ARCHITECTURE



[SECTOR 1: OPEN STANDARD VIDEO]

An accelerating push for HDZero and Walksnail to establish a modular, open-ecosystem hardware standard. The industry is moving to break away from closed-garden, proprietary digital video systems.

[SECTOR 2: ELRS 3.x & BEYOND]

LoRa modulation expanding far beyond simple RC control. The technical roadmap includes full MavLink data streaming, dual-band redundancy, and ultra-high bandwidth telemetry designed specifically to coordinate autonomous drone swarms.

[SECTOR 3: EDGE-AI PROCESSING]

The imminent transition from H7 MCUs to next-generation silicon. Architectures are shifting away from static PID loops toward real-time neural networks capable of predictive wind disturbance rejection and advanced sensor fusion directly on the flight controller.